



LEARNING INFRASTRUCTURE FOR THE NEXT BILLION CHILDREN

Programmable physical learning environments where children build, experiment and develop capabilities with shared AI.

Turn intelligence into capability.

Sam Quansah, founder · Cape Coast, Ghana · Pre-seed · September 2026

PROBLEM

ANSWERS ARE GETTING CHEAPER. CAPABILITY ISN'T.

AI makes explanations and finished work easy to get. The chance to investigate, build, get it wrong and try again still costs time, materials and skilled adults.

No equipment

A child in Cape Coast has few chances to experiment with real things.

AI everywhere

A child with plenty of AI can't explain or repeat the result when the help changes.

Unseen skill

A child shows real understanding through action, and no test records it.

The need is real: 70% of 10-year-olds in low- and middle-income countries can't read and understand a simple text.

World Bank, UNESCO and partners, 2022

INSIGHT

THERE IS NO RECORD OF WHAT CHILDREN CAN DO

The written record

Books, code, forums, images. Scraped at scale, and the foundation of today's AI.

The physical record

Real tasks with real objects: what a child did, with what help, and whether it held when the task changed. It happens in rooms and is almost never captured.

NBC builds the environments that develop it, and the records that show it: with consent, owned by families, and never sold.

SOLUTION

PROGRAMMABLE PHYSICAL LEARNING ENVIRONMENTS

Programmable materials

micro:bit, sensors and everyday materials whose roles and rules change, so one kit supports many worlds.

Learning Worlds

Designed learning experiences around real problems in the child's own world.

Shared AI

One AI shared by a centre helps facilitators prepare, translate and give feedback. No device per child.

Evidence of capability

What a child can explain, create and do when the task changes. Families can see and correct it.

Learning comes first. The experience has to be worth it for the child. Evidence supports the next learning decision, and a child's record is never the product.

WHAT A LEARNING WORLD IS

A DESIGNED EXPERIENCE, NOT A WORKSHEET

1 • Belong

A story and a real problem from the child's world. "Do I see my world here?"

2 • Investigate

Children build with programmable materials and ask AI good questions.

3 • Understand

Predict, test, explain, revise. "What is really going on underneath?"

4 • Apply

Build something for someone real, then show the skill in a new situation.

Each world ships with a **facilitator guide, materials list, AI prompts and short checks**. Printed cards and tokens are only the low-cost floor for rooms with nothing else.

Phases follow Algo Peers' Root Access pedagogy.

EXAMPLE WORLD • ALREADY RUN BY ALGO PEERS

FARM WORLD: TECHNOLOGY FOR FOOD SECURITY

What children did

Built AI pest detection, smart irrigation, plant growth and soil moisture monitors, and a vegetable classifier, using micro:bit, Teachable Machine, recycled materials and cocopeat. Delivered in person, in online sessions and through video.

51

learners aged 7–15

34

communities

53%

girls

7

working prototypes

Algo Peers module, documented at labs.algopeers.com. Participation and outputs, not a measured learning effect. NBC turns worlds like this into repeatable packages other facilitators can run.

WHY NOW

FOUR SHIFTS, ONE OPENING

Answers got cheap

AI costs keep falling.
Value moves to what
people can actually do.

Credentials are eroding

When work can be
generated, proof of
doing it yourself matters
more.

Computing became compulsory

Ghana's schools must
teach computing, and
most have few working
computers.

Shared AI is cheap enough

One phone or computer
can serve a whole
learning centre, offline
or online.

TRACTION

FIVE YEARS OF PRACTICE. NBC STARTS FROM IT.

Algo Peers, since 2021 · Cape Coast, Ghana

- 1,000+ children reached through our programmes
- Families pay for our after-school programmes
- \$93,511 in grants, including Global Affairs Canada
- 31 public schools assessed (11,214 students)
- Our own learning platform and educator training

NBC, started 2026

- Thesis, three Learning Worlds and a draft task format
- Website and classroom prototypes
- No NBC pilots or revenue yet
- Algo Peers' results don't prove NBC works. The pilot will test that

FIRST CUSTOMER

PROGRAMMES THAT ALREADY RUN HANDS-ON LEARNING

Independent operators of recurring hands-on programmes for children aged about 9–12, starting with reachable operators in Cape Coast.

ROLE	WHO
Uses it	Children in the programme
Runs it	A facilitator
Buys it	The programme owner. Parents may fund it through fees

Offer: NBC Learning Worlds. Reusable physical experiences, facilitator guidance and a short evidence summary that helps choose each child's next step.

MARKET

TAM, SAM, SOM

\$29–53B

TAM

Global after-school programmes, 2025 (industry estimates; methods vary). 1.4 billion students are enrolled worldwide.

\$99M

SAM

Ghana's ~27,400 primary and junior high schools × \$3,600 per site a year. Our first market.

\$3.2M

SOM, year 5

800 paying sites (about 3% of Ghana's schools) in our central scenario.

Site price = 240 participants × \$15 (the thesis's own illustration). Schools: Ghana education sector data (2020). Enrolment: UNESCO (2024). Market estimates: several research firms (2025).

BUSINESS MODEL

ONE CORE BUSINESS, OPTIONS LATER

Core

Learning service

A programme or site fee from operators. A family subscription through our platform is a separate test.

Option

Shared AI service

Useful AI work for operators and communities, only if they pay for it again and again.

Option

Creators and licensing

Licensed experience packages and set-up services, and later, verification.

Later, behind evidence gates: certification of schools and centres · pay-for-results contracts · physical-reasoning tests for AI labs, using paid adults only. **The learning business must stand alone,** and we never sell children's data.

UNIT ECONOMICS

HOW A SITE BREAKS EVEN

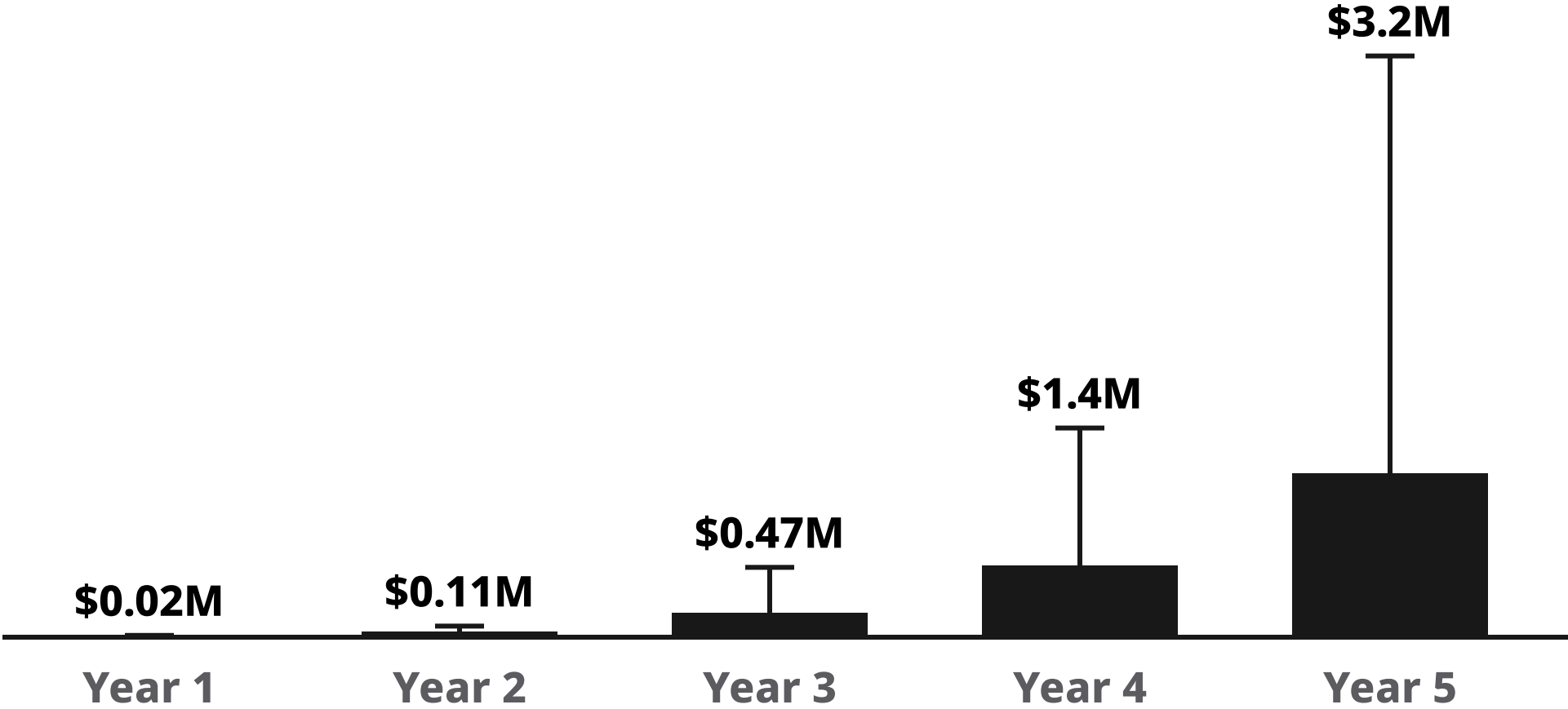
PARTICIPANTS A YEAR	COST PER PARTICIPANT
60	\$26
120	\$16
240	\$11

Example: \$1,200 fixed delivery cost a year plus \$6 per participant. At a price of \$15, a site covers these costs at **134 participants**.

Illustration only, not our price or forecast. Real prices come from the pilot's cost sheet. Next: scenarios built on these numbers.

PROJECTIONS

THREE SCENARIOS, FIVE YEARS



YEAR 5	SITES	REVENUE
Low	100	\$0.36M
Central	800	\$3.2M
High	2,500	\$11.3M

Bars: central scenario; lines: low to high. Revenue = sites × \$3,600, plus the options (0% low, 10% central, 25% high, from year 3). High adds a second country from year 3. Projections from stated assumptions, not forecasts. We rebuild them from pilot data.

WHERE THIS GOES

FROM A PRODUCT TO INFRASTRUCTURE

LAYER	WHAT IT IS	WHEN
Places	Kits, community centres and small schools, licensed to operators, never owned	Now
Learning Worlds	Designed experiences, written in an open format anyone can run	Now
Facilitators	A network of local people, helped by shared AI that works offline	Now
Records	What a child did, with what help, owned by the family	Year 1–2
The NBC Standard	Independent certification of worlds and centres: "the IB for the AI era"	Year 2–4
Exchange	Pay-for-results contracts, and physical-reasoning tests for AI with paid adults	Year 3+

WHY IT LASTS

WHAT A NEW AI MODEL CAN'T SHIP

People and places

Facilitators, kits and centres in real communities.

Trust

A standard run independently of any tech company.

Neutrality

We test and certify; we don't build the AI models.

Data that isn't online

Consented records of real physical learning, held for families.

An open format

Others build on it, so the whole ecosystem gains.

AI companies are our suppliers: every time AI gets cheaper, our cost per learner falls.

WHAT VENTURE SCALE REQUIRES

THE 10-YEAR AMBITION, IF NBC BECOMES THE STANDARD

LINE	WHAT HAS TO BE TRUE IN YEAR 10	REVENUE
Facilitator subscriptions	120,000 facilitators × \$350	\$42M
Certification	8,000 certified centres × \$5,000	\$40M
Pay-for-results	20M verified skills × \$2	\$40M
Small-school licences	2,500 schools × \$15,000	\$37.5M
AI testing (adults only)	10 contracts × \$8M, capped at half of revenue	\$80M
Kits and materials	At cost-plus	\$15M
Total	Range: \$94M to \$407M	\$254.5M

An ambition, not a projection. It needs every gate to pass, and a path from the high scenario into many countries.

COMPETITION

WHAT CHILDREN USE TODAY, AND WHAT WE MUST BEAT

	REAL HANDS-ON PRACTICE	WORKS WITHOUT A DEVICE PER CHILD	SHOWS WHAT A CHILD CAN DO
Tablet and AI tutor apps	No	No	Screen only
Robotics and coding kits	Yes	Costly	No
Exams and tests	No	Yes	Recall only
Good materials + a great teacher	Yes	Yes	Informal
NBC	Yes	Yes	Yes

Our real competitor is the last row. NBC has to make great hands-on learning cheaper to repeat and easier to run.

NEXT 90 DAYS

WHAT WE DO FIRST

Run the pilot

Eight weeks: 12–16 operator conversations, offers to 10 buyers, outside facilitators, delayed unfamiliar tasks.

Publish the format

Farm, Ramp and Plant worlds with guides, materials lists and AI prompts, under an open format.

Draft the Standard

NBC Standard v0.1 for worlds, reviewed by an advisory council.

Research only

A small AI testing study with paid adults and a public report. No children's data.

Pass lines, set before we start: 3 of 10 buyers pay · 2 outside facilitators deliver within support · results are readable · scorers agree · each site covers its direct costs.

MILESTONES AND GATES

PROVE IT, THEN NETWORK IT, THEN SET THE STANDARD

PHASE	WHEN	MILESTONES	GATE TO ADVANCE
Found	Months 0–3	Pilot starts; open format published; structure and data trust set up	Ethics approval; 1 outside facilitator
Prove	Months 3–12	2 sites (Cape Coast + a contrasting site); 20 outside facilitators; paid operators	Pilot pass lines met
Network	Months 12–30	500 facilitators in 3 countries; facilitator app; first certified worlds	Cost per learner falls across two cohorts
Standard	Years 3–5	8 countries; centres certified by independent assessors; first public contracts	A ministry or major network accepts the Standard

RISKS

WHAT COULD MAKE US WRONG

- **Substitution:** good materials and a great facilitator may do as well, more cheaply. Then we simplify to the part that helps.
- **Transfer** may not appear, or may cost too much to measure well.
- **Operators** may value the programme but not pay enough to cover delivery.
- **Trust:** automatic skill scoring could lose trust. Records stay human-readable, and NBC works without them.
- **Drift:** AI revenue could pull us off mission. We cap it at half of revenue, keep a firewall, and never use children's data.

TEAM

BUILT BY THE PEOPLE ALREADY IN THE ROOM

Sam Quansah

Founder

Founded Algo Peers in 2021. Ed.M. candidate, Harvard GSE. 2024 Mandela Washington Fellow. Former Google Africa Developer Scholarship mentor.

Vera Aninakwah

Learning Designer

Culturally responsive AI, computing and robotics for K-12. BBC micro:bit Champion and Africa Community Lead. Penn GSE scholarship recipient.

Nana Adwoa Nsiah

Learning Experience Designer

Leads our Learning Media Lab: animation and games for African children. micro:bit Champion. NYU Steinhardt scholarship recipient.

Hiring first: a learning scientist, and a technical lead for physical computing and offline software.

HOW CHILDREN WHO CAN'T PAY GET ACCESS

Sponsored schools

Companies and foundations fund Learning Worlds in public schools. Our non-profit delivers them.

Education authorities

License Learning Worlds and teacher training as computing becomes compulsory. After the pilot.

Community centres

Libraries and makerspaces host kits and a shared AI. Local government gives space; sponsors fund kit.

Universities

Co-design and independently evaluate our learning studies.

Two entities: Algo Peers Ltd builds and sells NBC. Algo Peers Impacts, our non-profit, delivers access. Sponsor money is ring-fenced, and children's data is never part of any deal.

THE ASK



RAISING A PRE-SEED ROUND

The pilot

The eight-week programme with external facilitators, delayed tests and paid buyers.

A second site

A contrasting setting, with different access to AI, for the same age group.

The team

A learning scientist and a technical lead, plus a properly powered follow-up study.

Use of funds (illustrative): people 35% · pilots and facilitators 20% · research, including the AI testing study 15% · legal, ethics and data trust 10% · product 10% · reserve 10%. Round details on request.

BUILT TO PROTECT CHILDREN'S DATA AND STAY NEUTRAL

Today

Algo Peers Ltd (commercial) builds and sells NBC. **Algo Peers Impacts** (non-profit) delivers access for children who can't pay.

Proposed as NBC grows

An **NBC Foundation** owns the open format and the Standard. An **NBC public-benefit company** runs the services. A **data trust** holds children's records for families. **Independent assessors** certify, so NBC never certifies itself.

Precedents: the Linux Foundation and Red Hat; UL; Coursera as a listed public-benefit company.

WHO BUILT LAYERS LIKE THIS

BENCHMARK	WHAT IT PROVES	LESSON FOR NBC
International Baccalaureate	A trusted network without owning schools	Trust takes years; start now
Red Hat and Linux	An open standard plus a commercial company	Open format; paid services
UL Solutions	Certification can be a large, listed company	Independent assessors
LEED	A voluntary standard reshaped an industry	Certify worlds first, then centres
ImageNet, ARC Prize	Benchmarks shape what AI labs build	Physical-reasoning tests, adults only
2U, Byju's	Debt-led expansion can sink edtech	Gate every spend on evidence

CHILDREN'S DATA IS NEVER THE PRODUCT

- No child's raw data leaves the data trust.
- No child's data trains or tests commercial AI models.
- Only group totals, above a minimum group size, and only with the trust's approval.
- Records separate what a child did from what we infer, and families can correct them.
- An annual independent public audit. A breach stops the work.

ONE FAMILY, ONE PROMISE

Architecture

NBC is the venture and the long-term infrastructure.
Algo Peers is the delivery partner and research base.
Learning Worlds is the offer. **Papa Algo** is the character children meet.

Message levels

Master: Learning infrastructure for the next billion children. **Vision:** Turn intelligence into capability. **Offer:** Change one thing. **Proof:** What we know. What we're testing.

Experience

Learning Worlds, kits, materials

Intelligence

AI, always with human review

Evidence

What a child did, and how

Capability

Shown again when the task changes. Never a grade

Action

The child's move, and one button per view

Built on the Algo Peers brand system: seven colours, expanded uppercase titles, single-colour logos, flat blocks. Voice: inspirational but grounded, with the place named and hard numbers.